Arista Access Router Hardware/Environmental Fault Failure Domain Resolution (HEB-ACC-25 Hebbal Edge)

Node: HEB-ACC-25 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_3573

Description

This Standard Operating Procedure (SOP) outlines the step-by-step approach to troubleshoot and resolve hardware/environmental fault failures on the Arista Access Router (HEB-ACC-25) located at the Hebbal Edge site. The failure domain in question is characterized by ASIC memory parity errors, line card crashes, or loss of power supply redundancy, culminating in a Major (ALARM_3573) alarm. The root cause of this failure is often linked to the inherent vulnerabilities of modern networking hardware, particularly Arista's hardware design, where the reliance on ASICs (Application-Specific Integrated Circuits) for accelerated packet processing introduces the risk of memory parity errors and subsequent line card crashes. Furthermore, the use of redundant power supplies (N+1 redundancy) in the Hebbal Edge site may not be sufficient to mitigate the impact of a single power supply failure. Additionally, environmental factors such as temperature fluctuations, humidity, or power quality issues may also contribute to the failure. Therefore, this SOP aims to provide a structured approach to identifying and resolving the root cause of the failure, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Ensure that the Hebbal Edge site has been secured and access restricted to authorized personnel only.

- Verify that the redundant power supplies are functioning correctly and that the line cards are online.

- Check the environmental conditions within the site, including temperature and humidity levels, to ensure they are within the recommended operating range for the Arista Access Router.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System Resources: Memory: MB, CPU: %, Description: This command displays the current system resource utilization, including memory and CPU usage. This information can help identify whether the ASIC memory parity error is causing a significant impact on system resources.
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```
Command: show system redundancy, Expected Output: Redundancy Status: Power Supplies: N+1, Line Cards: Online, Description: This command displays the current redundancy status of the power supplies and line cards. This information can help determine whether the power supply redundancy loss is the root cause of the failure.

Command: show system environment, Expected Output: Environmental Conditions: Temperature: °C, Humidity: %, Description: This command displays the current environmental conditions within the site, including temperature and humidity levels. This information can help identify whether environmental factors are contributing to the failure.
```

Resolution Steps

Step 1: Step: Replace faulty power supply, Command: replace system power-supply , Description: To resolve the power supply redundancy loss, replace the faulty power supply with a new one. This can be done by executing the 'replace system power-supply' command, followed by the number of the power supply to replace.

Step 2: Step: Reset line card, Command: reset system line-card , Description: To resolve the line card crash, reset the affected line card by executing the 'reset system line-card' command, followed by the number of the line card to reset.

Step 3: Step: Clear ASIC memory parity error, Command: clear system asic-memory-parity-error, Description: To resolve the ASIC memory parity error, clear the error by executing the 'clear system asic-memory-parity-error' command. This will reset the affected ASIC and clear the error.

Post-Check Verification

```
Command: show system resource, Expected Output: System Resources: Memory: MB, CPU: %, Description: This command displays the current system resource utilization, including memory and CPU usage. Verify that the resource utilization has returned to normal after the resolution steps.

Command: show system redundancy, Expected Output: Redundancy Status: Power Supplies: N+1, Line Cards: Online, Description: This command displays the current redundancy status of the power supplies and line cards. Verify that the redundancy status has been restored to normal after the resolution steps.

Command: show system environment, Expected Output: Environmental Conditions: Temperature: °C, Humidity: %, Description: This command displays the current environmental conditions within the site, including temperature and humidity levels. Verify that the environmental conditions have not changed after the resolution steps.
```

Roll-back Procedures

- Step: Restore original power supply, Command: insert system power-supply , Description: If the resolution step to replace the faulty power supply fails, restore the original power supply by executing the 'insert system power-supply' command, followed by the number of the original power supply.

- Step: Restore original line card, Command: install system line-card , Description: If the resolution step to reset the line card fails, restore the original line card by executing the 'install system line-card' command, followed by the number of the original line card.

- Step: Re-enable ASIC memory parity error, Command: enable system asic-memory-parity-error, Description: If the resolution step to clear the ASIC memory parity error fails, re-enable the error by executing the 'enable system asic-memory-parity-error' command.